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95%

TRANSPORT PHENOMENA II, CHE 312

QUIZ 2

CLOSED BOOK, TIME: 20 MINUTES

A cold liquid film is flowing down a vertical solid wall with the velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right]$$

The liquid film, initially at a uniform temperature T_1 comes into contact with a warm gas that maintains the gas-liquid interface temperature at T_0 .

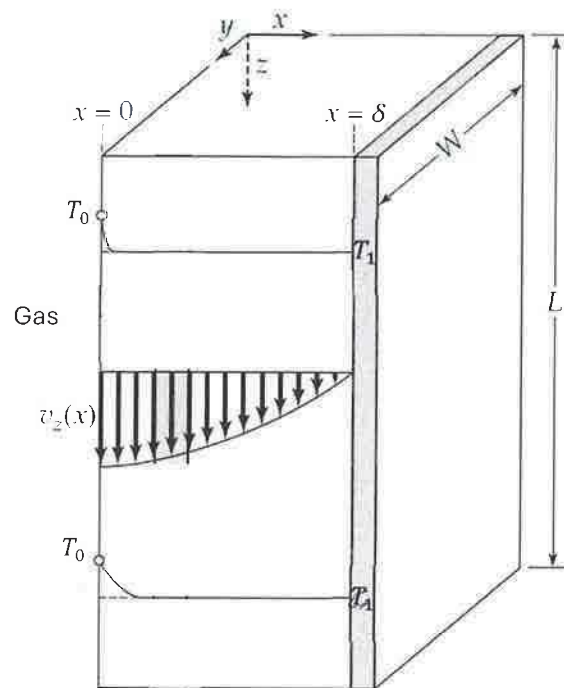
Postulates:

- Liquid has constant density ρ , heat capacity c_p , and thermal conductivity k .
- Short contact times, so the liquid temperature changes appreciably only in the immediate vicinity of the gas interface.
- For $\frac{x}{\delta} \ll 1$ assume $v_z \approx v_{z,max}$

1- Simplify the equation of energy to derive a partial differential equation (PDE) for the steady-state dimensionless temperature $\theta = \frac{T - T_1}{T_0 - T_1}$ as a function of x and z .

2- Convert the above PDE to an ordinary differential equation for θ as a function of $\eta = \frac{x}{\sqrt{4\beta z}}$, where $\beta = \frac{k}{\rho c_p v_{z,max}}$.

3- Write down two boundary conditions for $\theta = \theta(\eta)$.



§B.9 THE EQUATION OF ENERGY FOR PURE NEWTONIAN FLUIDS WITH CONSTANT^a ρ AND k

$$[\rho \hat{C}_p DT/Dt = k \nabla^2 T + \mu \Phi_v]$$

Cartesian coordinates (x, y, z) :

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (\text{B.9-1})^b$$

$$\hat{C}_p \rho v_z \frac{\partial u}{\partial z} = k \frac{\partial^2 u}{\partial x^2} \quad \frac{\partial u}{\partial z} = 0 \Rightarrow \frac{\partial^2 T}{\partial z^2} = 0$$

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1. Energy equation

$$p c_p v_2 \frac{\partial T}{\partial z} = k \left[\frac{\partial T^2}{\partial x^2} \right] \checkmark$$

1) good

3) good

2) minor alg. error

-5

$$2. p c_p \frac{\partial T}{\partial z} = \frac{k}{p c_p v_2} \left[\frac{\partial T^2}{\partial x^2} \right]$$

$$\Rightarrow p c_p v_{2, \max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right] \frac{\partial T}{\partial z} = k \left[\frac{\partial T^2}{\partial x^2} \right]$$

$$\Rightarrow \left[1 - \left(\frac{x}{\delta} \right)^2 \right] \frac{\partial T}{\partial z} = B \left[\frac{\partial T^2}{\partial x^2} \right] \checkmark$$

$$\theta(T_0 - T_1) = T - T_1 \Rightarrow \frac{\partial T}{\partial z} = \frac{\partial \theta}{\partial z} \cdot (T_0 - T_1)$$

$$\frac{\partial T}{\partial z} = \frac{\partial \theta}{\partial n} \cdot \frac{\partial n}{\partial z} \cdot (T_0 - T_1)$$

$$\frac{\partial n}{\partial z} = \frac{\partial (x \cdot (4Bz)^{-1/2})}{\partial z} = \frac{-1}{2} \cdot 2^{-1/2-1} = \frac{-x}{\sqrt{4B}} \cdot \frac{1}{2z^{3/2}}$$

$$\Rightarrow \frac{\partial T}{\partial z} = \frac{\partial \theta}{\partial n} \cdot \frac{x}{\sqrt{4B}} \cdot \frac{1}{2z} \cdot (T_0 - T_1)$$

$$\frac{\partial T^2}{\partial x^2} \Rightarrow \frac{\partial T}{\partial x} = \frac{\partial \theta}{\partial n} \cdot \frac{\partial n}{\partial x} \cdot (T_0 - T_1)$$

$$\Rightarrow \frac{\partial \theta}{\partial n} \cdot \frac{1}{\sqrt{4Bz}} (T_0 - T_1)$$

$$\Rightarrow \frac{\partial T^2}{\partial x^2} = \frac{\partial^2 \theta}{\partial n^2} \cdot \frac{4x}{\sqrt{4Bz}} (T_0 - T_1) \checkmark$$

$$\Rightarrow \left(1 - \left(\frac{x}{\delta} \right)^2 \right) \frac{\partial \theta}{\partial n} \cdot \frac{x}{\sqrt{4B}} \cdot \frac{1}{2z} = B \cdot \frac{\partial^2 \theta}{\partial n^2} \cdot \frac{4x}{\sqrt{4Bz}}$$

$$\Rightarrow \left(1 - \left(\frac{x}{\delta} \right)^2 \right) \frac{\partial \theta}{\partial n} \cdot \frac{1}{2z} = B \frac{\partial^2 \theta}{\partial n^2} \Rightarrow \left(1 - \left(\frac{x}{\delta} \right)^2 \right) \frac{\partial \theta}{\partial n} \cdot \frac{1}{2z} = B \frac{\partial^2 \theta}{\partial n^2}$$

since $v_2 \approx v_{2, \max}$

$$3. \theta(r=1) = T_1$$

$$\checkmark \theta(r=0) = T_0$$

$$\theta(r=1) = T_1$$

$$\frac{\partial u}{\partial z} \Big|_{x=0} = 0$$

$$\Rightarrow \theta(r=0) = T_0$$

$$\checkmark \theta(r=\frac{\delta}{\sqrt{4Bz}}) = T_1$$

$$\Rightarrow \theta(n-\delta) = T_1$$

$$\frac{x}{\sqrt{4Bz}^{1/2}}$$

$$\frac{\partial \theta}{\partial n} \cdot \frac{1}{2z} = B \frac{\partial^2 \theta}{\partial n^2}$$

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